

PROFESSIONAL INSTALLATION MANUAL

Week 3 - Enterprise Server Deployment

ITEP 414 - System Administration and Maintenance

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1. Introduction

This manual provides a repeatable procedure for deploying and validating the Week 3 Ubuntu Server virtual machine. It is written so another administrator can follow the process without additional instructions.

2. Ubuntu Server Installation Guide

1. Create the Virtual Machine

Create a VM named Ubuntu-Server-Week03 with at least 4 GB RAM, 2 virtual CPUs, a 40 GB virtual disk, and NAT networking. Attach the Ubuntu Server LTS ISO.

2. Start the Installer

Boot the VM from the ISO and select English.

3. Keyboard Layout

Select the appropriate layout, normally English (US).

4. Network Configuration

Accept DHCP unless your instructor requires a static address. Record the assigned IP address.

5. Storage

Choose Guided - Use Entire Disk. Record the disk size, partition layout, and filesystem shown by the installer before confirming.

6. Profile / Hostname

Set the server hostname to server01. Create a non-root administrative account using your own name and username, with a strong password.

7. SSH

Enable Install OpenSSH Server so the server can be administered remotely.

8. Additional Packages

Do not select additional packages unless instructed.

9. Complete Installation

Finish installation, reboot the VM, and remove/eject the installation ISO when prompted.

3. Initial Server Configuration

Configuration	Required Value / Action
Hostname	server01
Administrative user	Non-root user with sudo privileges
Networking	DHCP unless otherwise instructed; record assigned IP
Storage	Guided - Use Entire Disk; document layout/filesystem
SSH	Install OpenSSH Server
Packages	No additional packages unless instructed

4. Server Verification Procedures

Login

Log in using the non-root administrative account.

Hostname

Run: hostname. Expected hostname: server01.

IP Address

Run: ip addr. Capture the active interface and assigned IP.

Internet Connectivity

Run: ping -c 4 google.com. Capture the successful replies.

System Update

Run: sudo apt update, then sudo apt upgrade -y. Capture successful update/upgrade output.

SSH Service

Run: systemctl status ssh. It should show active (running).

5. Troubleshooting Guide

Problem	Checks / Resolution
VM does not boot installer	Confirm ISO is attached to the virtual optical drive and optical media is before the virtual disk in boot order.
No IP address	Check VM network adapter is enabled and attached to NAT/Bridged; run ip addr; renew/reboot if necessary.
ping google.com fails	Check IP address, default route, DNS resolution, and VM network mode. Test an IP address separately to

	distinguish DNS from connectivity.
apt update fails	Check internet/DNS connectivity and retry. Read the repository error before changing configuration.
SSH service not running	Run systemctl status ssh. If OpenSSH was not installed, install openssh-server, then enable/start the service.
Cannot SSH from host	Confirm the service is active, correct IP is used, firewall/network mode permits access, and the VM is reachable from the host.

6. Best Practices

- Use a non-root administrative account and sudo.
- Apply security updates after installation.
- Use strong unique credentials and SSH keys where appropriate.
- Document hostname, addressing, storage, and service configuration.
- Back up important configuration and data before major changes.
- Verify services after changes rather than assuming they started successfully.
- Keep screenshots and command output as deployment evidence.

7. Windows Server Installation Summary

Install Windows Server Evaluation in a separate virtual machine. Assign a computer name, create the Administrator password, log in successfully, and capture the required installation evidence. Insert your actual screenshots below.

[INSERT YOUR SCREENSHOT HERE: Windows Server installation/setup]

Recommended filename: screenshots/07-windows-installation.png

[INSERT YOUR SCREENSHOT HERE: Windows Server computer name]

Recommended filename: screenshots/08-windows-computer-name.png

[INSERT YOUR SCREENSHOT HERE: Windows Server successful login]

Recommended filename: screenshots/09-windows-login.png

8. Enterprise OS Comparison

Category	Windows Server	Ubuntu Server	Rocky Linux
Licensing	Commercial Microsoft licensing; evaluation edition is time-limited.	Open-source; free to use, with optional paid Ubuntu Pro/support.	Open-source community Enterprise Linux; no license fee for the OS.
User Interface	Server Core or Desktop Experience GUI.	Primarily command-line on Server; optional GUI packages.	Primarily command-line for servers; desktop environments are available.
Package Management	Windows Update, PowerShell, roles/features, package tools.	APT / dpkg with Ubuntu repositories.	DNF / RPM with Enterprise Linux repositories.
Security	Active Directory ecosystem,	AppArmor, UFW/nftables, SSH,	SELinux, firewalld/nftables, SSH,

	Defender/security baselines, Windows Firewall, BitLocker and enterprise management options.	unattended/security updates, strong Linux permissions.	Enterprise Linux security model.
Performance	Strong for Microsoft workloads and Windows-native enterprise applications; GUI edition has additional overhead.	Efficient server footprint; common for web, database, container, and cloud workloads.	Stable enterprise-oriented Linux platform suited to long-running server workloads.
Typical Enterprise Use	Active Directory, Windows file services, IIS, .NET, Microsoft-centric infrastructure.	Web/database servers, cloud workloads, DevOps, containers, general Linux infrastructure.	RHEL-compatible enterprise servers, hosting, databases, infrastructure requiring Enterprise Linux compatibility.
Advantages	Deep Microsoft integration, mature GUI/admin ecosystem, AD and Group Policy.	No OS license fee, large community, strong cloud/container ecosystem, LTS releases.	Enterprise Linux compatibility, long-term stability focus, community-driven and open-source.
Disadvantages	Licensing cost/complexity; Windows-specific administration overhead.	Some proprietary enterprise applications may not support Linux; CLI skills often required.	Smaller desktop/general-user ecosystem; some commercial vendors certify RHEL before Rocky Linux.

9. References

Week 3 Portfolio Project - ITEP 414 module.

Ubuntu Server documentation - <https://help.ubuntu.com/>

Microsoft Windows Server documentation - <https://learn.microsoft.com/windows-server/>

Rocky Linux documentation - <https://docs.rockylinux.org/>